Preview: y2002_q21 (v1)

Reach a target with + – × ÷ using each card once; know when it is impossible

Part A: Each question is worth 3 points

1. The digits from 1 to 9 are written on 9 cards. Alex has 2, 6 and 8; Omar has 4, 3 and 1; Lily has 9, 7 and 5. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who *cannot* get 10 as the result?

A. Alex ***B.*** Omar ***C.*** Lily
D. All can get 10 ***E.*** Alex and Omar

2. The digits from 1 to 9 are written on 9 cards. Tom has 3, 5 and 9; Omar has 2, 6 and 8; Martha has 1, 4 and 7. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who *cannot* get 20 as the result?

A. Tom ***B.*** Omar ***C.*** Martha
D. All can get 20 ***E.*** Tom and Martha

3. Omar has the cards 2, 5 and 6. Omar must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Omar get?

A. 19 ***B.*** 15 ***C.*** 14 ***D.*** 12 ***E.*** 11

4. Mia has the cards 1, 5, 7 and 9. Mia must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Mia get?

A. 55 ***B.*** 54 ***C.*** 49 ***D.*** 48 ***E.*** 42

Part B: Each question is worth 4 points

5. Omar has the cards 2, 7 and 9. Omar must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Omar *not* get?

A. 32 ***B.*** 24 ***C.*** 23 ***D.*** 18 ***E.*** 14

6. Tom has the cards 1, 4, 5 and 7. Tom must use every card exactly once, with +, –, ×, ÷ and brackets.
Which number can Tom *not* get?

A. 41 ***B.*** 38 ***C.*** 37 ***D.*** 34 ***E.*** 33

7. Omar has three cards with different digits. Two of them are 9 and 3; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Omar gets 26.
Which of these digits can be on the hidden card?

A. 1 ***B.*** 2 ***C.*** 4 ***D.*** 6 ***E.*** 8

8. Alex has three cards with different digits. Two of them are 6 and 7; the third is face down. Using each card exactly once, with +, –, ×, ÷ and brackets, Alex gets 37.
Which of these digits can be on the hidden card?

A. 1 ***B.*** 4 ***C.*** 5 ***D.*** 8 ***E.*** 9

Part C: Each question is worth 5 points

9. Martha has the cards 1, 4 and 8. Using every card exactly once, with +, –, ×, ÷ and brackets, Martha wants to get the biggest possible number.
What is the biggest number Martha can get?

A. 48 ***B.*** 45 ***C.*** 44 ***D.*** 41 ***E.*** 40

10. Martha has the cards 1, 4, 8 and 9. Using every card exactly once, with +, –, ×, ÷ and brackets, Martha wants to get the biggest possible number.
What is the biggest number Martha can get?

A. 443 ***B.*** 398 ***C.*** 360 ***D.*** 288 ***E.*** 22

Answer Key

1.	**B**	**2.**	**E**	**3.**	**B**	**4.**	**C**	**5.**	**B**	**6.**	**A**
7.	**A**	**8.**	**C**	**9.**	**E**	**10.**	**C**				

Solutions

1. B (Omar) — Alex: 2 × 8 – 6 = 10; Lily: 5 × (9 – 7) = 10. Omar: with 4, 3 and 1 nothing gives 10 (the closest are 9 = (4 – 1) × 3 and 11 = 3 × 4 – 1). [2002 Q21 · who_cannot]

2. E (Tom and Martha) — Omar: 2 × 6 + 8 = 20. Tom: with 3, 5 and 9 nothing gives 20 (the closest are 18 = (5 – 3) × 9 and 22 = 3 × 9 – 5); Martha: with 1, 4 and 7 nothing gives 20 (the closest are 12 = 1 + 4 + 7 and 21 = (4 – 1) × 7). [2002 Q21 · who_cannot]

3. B (15) — 5 × 6 ÷ 2 = 15. None of the other numbers can be made. [2002 Q21 · which_can]

4. C (49) — (1 + 7) × 5 + 9 = 49. None of the other numbers can be made. [2002 Q21 · which_can]

5. B (24) — 7 + 9 – 2 = 14; 2 + 7 + 9 = 18; 2 × 7 + 9 = 23; 2 × (7 + 9) = 32. But with 2, 7 and 9 nothing gives 24 (the closest are 23 = 2 × 7 + 9 and 25 = 2 × 9 + 7). [2002 Q21 · which_cannot]

6. A (41) — 1 × 5 + 4 × 7 = 33; 1 + 4 × 7 + 5 = 34; (1 + 7) × 4 + 5 = 37; 4 + 5 × 7 – 1 = 38. But with 1, 4, 5 and 7 nothing gives 41 (the closest are 40 = 1 + 4 + 5 × 7 and 44 = (5 + 7 – 1) × 4). [2002 Q21 · which_cannot]

7. A (1) — 3 × 9 – 1 = 26. The other digits do not work with 9 and 3. [2002 Q21 · hidden_card]

8. C (5) — 6 × 7 – 5 = 37. The other digits do not work with 6 and 7. [2002 Q21 · hidden_card]

9. E (40) — (1 + 4) × 8 = 40. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

10. C (360) — (1 + 4) × 8 × 9 = 360. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]